NAME Solution

ChE 210 Quiz #1 13 September 2019

In class, we discussed fitting human blood viscosity data using the Sisko Model for a shear-rate-dependent viscosity $\eta(\dot{\gamma})$: $\eta - \eta_\infty = m\,\dot{\gamma}^{n-1}$
where η_∞, m, and n are three adjustable constants. It is know that for human blood that $n = 0.5$ is a good approximation.

(a) Assuming that $n = 0.5$, how can the Sisko model be linearized so that linear least squares can be used to determine the remaining two constants? Show clearly what you would plot as x on the horizontal axis and y on the vertical axis.

(b) Use the data provided to calculate the remaining two parameters using horizontal offsets. Be sure to give the correct units.

For horizontal offsets: slope $a = \frac{S_{yy} - S_y^2}{S_{xy} - S_x S_y}$ and intercept $b = \frac{S_{xy} S_y - S_{yy} S_x}{S_{xy} - S_x S_y}$.

$\frac{1}{\sqrt{\dot{\gamma}}}$	Shear Rate $\dot{\gamma}$ (s^{-1})	Viscosity η (mPa·s)
1.40	0.51	39.15
0.864	1.34	22.18
0.354	7.97	9.51
0.105	90.73	5.77
0.0363	759.07	4.23
S_x 0.552		S_y 16.17

$S_{yy} = 433.26$ $S_{xy} = 15.62$

(a) $\eta = m\left[\frac{1}{\sqrt{\dot{\gamma}}}\right] + \eta_\infty$

$y = a\,x + b$

$\therefore\ \eta \rightarrow y$ and $\frac{1}{\sqrt{\dot{\gamma}}} \rightarrow x$

$m \rightarrow a$ and $\eta_\infty \rightarrow b$

Plot $\frac{1}{\sqrt{\dot{\gamma}}}$ vs. η

(b) $a = \frac{433.26 - (16.17)^2}{15.62 - (0.552)(16.17)} = 25.66\ \frac{\text{mPa·s}}{\sqrt{s}} = 25.66\ \text{mPa}\sqrt{s}$

$b = \frac{(15.62)(16.17) - (433.26)(0.552)}{15.62 - (0.552)(16.17)} = 2.604\ \text{mPa·s}$

Note: without round off error

$a = 25.58\ \text{mPa}\sqrt{s}$

$b = 2.031\ \text{mPa·s}$